a Aim 1: research held to the frameworks

Question and its gate set by a person, sealed

Hypothesis sealed; jobs run through Beethoven; code checked by beethoven-qa

REFORMS record kept; does the gate pass?

It passes: a pull request that a person merges

Symphony: one agent holds each task; agents work in parallel

Not yet: the result, null or not, is kept; the agent revises its hypothesis

Scored by a hidden check, with and without the checks

b Aim 2: the trainee's agent

claude.ai in the trainee's browser, with the Beethoven extension

a chat

one click: a Code session

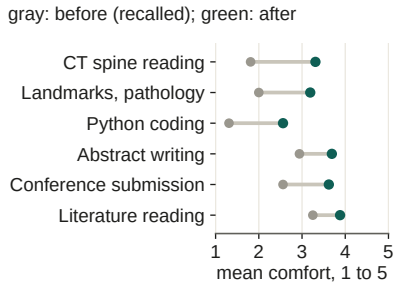
Claude servers: the model

WSU grid: the trainee's Claude Code worker, a Slurm job

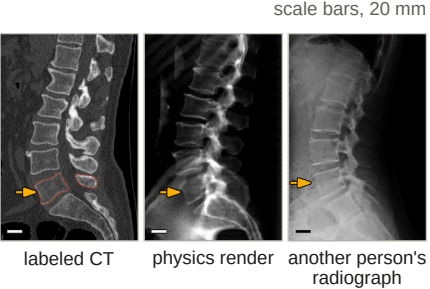
built to send GPU jobs

H200 GPUs, 4 nodes of 4

c 2026 cohort, n = 16



d Aim 3: from labeled CT to X-ray



e Aim 3: the spine RAG-LLM

The image, read by the detectors and the level model

Scopus and ScienceDirect spine literature, indexed

reading

passages

Multimodal RAG-LLM on the WSU grid, fine-tuned on public images, then on images with charts under the IRB

Answer: level, finding, verbatim quotations

graded against OpenEvidence

f Held-out results on a public C-arm

